Sequence and Series

$a_n = 2n-1$
$a_n = n^2$
> Last element of a Sequence.

$a = 1, 3, 5, 7 \ldots\ldots$ $(2n-1)$
$a = 1, 4, 9, 16 \ldots$
> Infinite Sequence

Convergency and Divergency:

Find nth element (a_n) Last element

$\lim_{n \to p}$ → Defined (convergent)
→ Undefined (Divergent)

e.g

$a = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5} \ldots\ldots$

$a_n = \frac{n}{n+1}$

$\lim_{n \to \infty} \left(\frac{n}{n+1} \right)$

$\lim_{n \to \infty} \left(\frac{n}{n\left(1 + \frac{1}{n}\right)} \right) \rightarrow \lim_{n \to \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)$

$$\frac{1}{1+\frac{1}{\infty}} = \frac{1}{1+0} = (1)$$ Convergent

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$a = 1, 4, 9, 16$

$a_n = n^2$

$\lim_{n \to \infty} (\infty)^2 = \infty$ Divergent

Sandwitch theorem for Sequences.

$a_n = \frac{1}{n^2} \sin(n)$ ← Example

$f(n) \leq g(n) \leq h(n)$

$\lim_{n \to c} f(n) = L$

$\lim_{n \to c} h(n) = L$ then

$$\lim_{n \to c} g(n) = L$$

$$a_n = \frac{1}{n^2}\sin(n)$$

$$\frac{-1}{n^2} \le \frac{\sin(n)}{n^2} \le \frac{1}{n^2}$$

$$\lim_{n\to\infty}\left(\frac{-1}{n^2}\right) = \frac{-1}{(\infty)^2} = 0$$

$$\lim_{n\to\infty}\left[\frac{1}{n^2}\right] = \frac{-1}{(\infty)^2} = 0$$

hence

$$\lim_{n\to\infty}\frac{\sin(\infty)}{(\infty)^2} = \boxed{0}$$